HC 6001: Scattering by a Central Potential; Hamiltonian and Hamilton's Equations of Motion

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1 Scattering by a Central Potential

1.1 Kinematics and Definitions

$$\text{Incident energy: } E = \frac{m}{2}v_0^2$$
$$\text{Incident velocity: } v_0 = \sqrt{2E/m}$$
$$\text{Angular momentum: } \ell = smv_0 = s\sqrt{2mE}, \quad s = \frac{\ell}{\sqrt{2mE}}$$

Here s is the impact parameter: particles are incident with impact parameter between s and $s + ds$, and travel along the incoming direction until they reach the apsidal point (distance of closest approach) and emerge scattered through angle θ relative to the incident direction. By azimuthal symmetry the scattering does not depend on the azimuthal angle φ; only θ (and E) matter, so the scattered intensity distribution is written $\sigma(\theta, E)$.

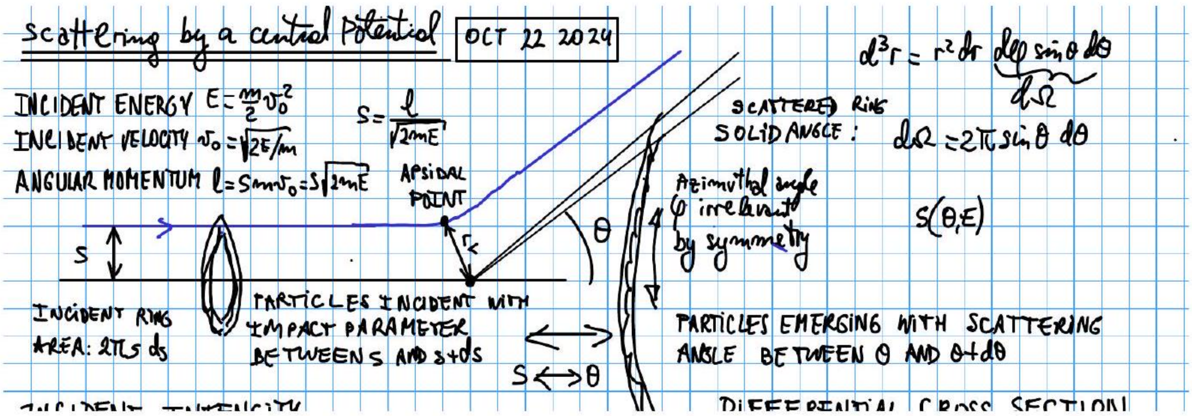

Figure 1: Incident-beam geometry: impact parameter s, apsidal point, and scattering angle θ. Volume element $d^3r = r^2 dr \, d\Omega$ with $d\Omega = \sin\theta \, d\theta \, d\varphi$.

The solid angle into which particles are scattered is $d\Omega = 2\pi \sin\theta \, d\theta$ (azimuthal integration already performed, using the symmetry above).

1.2 Differential and Total Cross Section (General Definition)

The incident ring of particles with impact parameter between s and $s + ds$ has area $2\pi s \, ds$. Define the incident intensity

$$I = \frac{\#\text{ particles incident per unit time}}{\text{incident beam cross-section area}}$$

and the differential cross section $\sigma(\Omega)$ via

$$\sigma(\Omega)\, d\Omega = \frac{\#\text{ particles scattered into } d\Omega \text{ per unit time}}{\text{incident intensity}}$$

Particle-number conservation between the incident ring and the emerging angular range gives

$I(2\pi s)|ds| = \#$ particles between s, $s{+}ds = \#$ particles emerging between θ, $\theta{+}d\theta = I\sigma(\theta)\, 2\pi \sin\theta\, |d\theta|$.

Therefore

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right| \tag{1}$$

and the total cross section is defined as

$$\sigma_T = \int d\Omega\, \sigma(\Omega) \tag{2}$$

1.3 Relating θ and ψ: Repulsive and Attractive Cases

Let ψ be the angle swept from the incident asymptote to the apsidal point. Two geometric cases arise depending on whether the interaction is repulsive or attractive:

Repulsive case:

$$\pi = \theta + 2\psi \quad \Longrightarrow \quad \theta = \pi - 2\psi, \quad \psi = \frac{\pi - \theta}{2}$$

Attractive case:

$$\pi = 2\psi - \theta \quad \Longrightarrow \quad \theta = 2\psi - \pi, \quad \psi = \frac{\pi + \theta}{2}$$

In all cases:

$$\theta = |\pi - 2\psi| \tag{3}$$

1.4 Computing $S(\theta, E)$: the Universal Method

Starting from energy conservation with the centrifugal term,

$$E = \frac{m}{2}\dot{r}^2 + V(r) + \frac{\ell^2}{2mr^2} \quad \Longrightarrow \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{\ell^2}{2mr^2} \right) \right]^{1/2}, \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

From $\ell = mr^2\dot{\theta}$,

$$d\theta = \dot{\theta}\, dt = \frac{\ell}{mr^2}\, dt = \frac{\ell}{mr^2} \cdot \frac{dr}{\dot{r}} = \frac{\ell}{m}\sqrt{\frac{m}{2E}} \frac{r^{-2}\, dr}{\left[1 - \frac{V(r)}{E} - \frac{\ell^2}{2mEr^2}\right]^{1/2}}$$

where $\ell/\sqrt{2mE} = s$ and $\ell^2/(2mE) = s^2$ (both from the definition of s above), so

$$d\theta = \frac{s\, r^{-2}\, dr}{\left[1 - \frac{V(r)}{E} - \frac{s^2}{r^2}\right]^{1/2}} \tag{4}$$

Integrating from the apsidal point ($r = r_m, \theta = \theta_m$) out to $r \to \infty$ ($\theta \to \pi$):

$$\psi = \int_0^\psi d\psi = \int_{\theta_m}^\pi d\theta = \int_{r_m}^\infty \frac{s\, r^{-2}\, dr}{\left[1 - \frac{V(r)}{E} - \frac{s^2}{r^2}\right]^{1/2}}$$

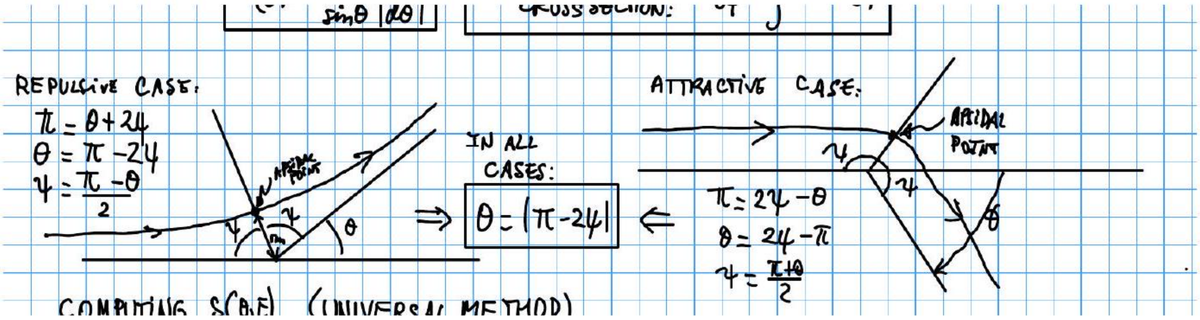

Figure 2: Repulsive-case (left) and attractive-case (right) scattering geometry, relating the apsidal-point sweep angle ψ to the scattering angle θ; in all cases $\theta = |\pi - 2\psi|$.

Substituting $u = 1/r,\ du = -r^{-2}\,dr,\ u_m = 1/r_m$, and using $\theta = |\pi - 2\psi|$:

$$\theta(s) = \left| \pi - 2\int_0^{u_m} \frac{s\,du}{\left[1 - \frac{V(1/u)}{E} - s^2u^2\right]^{1/2}} \right| \tag{5}$$

From here one computes $\left|\frac{ds}{d\theta}\right| = 1/\left|\frac{d\theta}{ds}\right|$ and uses it to get $\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right|$.

Remark 1. Up to here, all results are valid for any central potential $V(r)$.

2 Coulomb (Rutherford) Scattering

2.1 Setup: Orbit Equation and Eccentricity

For the Coulomb force between charges Ze and $Z'e$,

$$f = \frac{ZZ'e^2}{r^2}, \quad V(r) = -\frac{k}{r}, \quad k = -ZZ'e^2.$$

$$\begin{array}{lll} E > 0 & \text{hyperbolic} & \\ E = 0 & \text{parabolic} & (\text{only for } V = -k/r\ ) \\ E < 0 & \text{elliptic} & \end{array}$$

For scattering we take $E > 0$ (hyperbolic orbit). The orbit equation is

$$\frac{1}{r} = C_1\left[1 + \epsilon\cos\left(\tilde\theta - \tilde\theta'\right)\right], \quad C_1 = \frac{mk}{\ell^2} = -\frac{m}{\ell^2}ZZ'e^2$$

with eccentricity

$$\epsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \frac{2E\ell^2}{m\left(ZZ'e^2\right)^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1$$

using $\ell = smv_0 = s\sqrt{2mE}$, so $\ell^2 = 2mEs^2$ and $2E\ell^2/\left(mk^2\right) = (2sE/k)^2$.

Two situations arise:

i) Repulsive case: $ZZ' > 0 \Rightarrow k = -ZZ'e^2 < 0$

i) Attractive case: $ZZ' < 0 \Rightarrow k = -ZZ'e^2 > 0$ (also gravity, $k = Gm_1m_2$)

2.2 Repulsive Case

Here $k < 0$, so $C_1 = mk/\ell^2 < 0$. Physical orbits require $1/r > 0$, i.e. $C_1\left[1+\epsilon\cos\left(\tilde\theta-\tilde\theta'\right)\right] > 0 \Leftrightarrow \cos\left(\tilde\theta-\tilde\theta'\right) < -1/\epsilon$: we need the cosine to be negative and larger in magnitude than $1/\epsilon$.

$\tilde\theta = \tilde\theta'$ is not a physical point, since $\cos\left(\tilde\theta'-\tilde\theta'\right) = \cos 0 = 1$ would give $1/r = C_1(1+\epsilon) < 0$ (unphysical). So $\tilde\theta = \tilde\theta'$ lies in the middle of the inaccessible region of the hyperbola, and $\tilde\theta = \tilde\theta' \pm \pi$ is the periapsis; choose $\tilde\theta' = \pi$:

$$\cos\left(\tilde\theta-\tilde\theta'\right) = \cos(\tilde\theta-\pi) = \cos\tilde\theta\cos\pi + \sin\tilde\theta\sin\pi = -\cos\tilde\theta$$

so

$$\frac{1}{r} = \left(-\frac{mk}{\ell^2}\right)\left[-1-\epsilon\cos\left(\tilde\theta-\tilde\theta'\right)\right] = \frac{mZZ'e^2}{\ell^2}[\epsilon\cos\tilde\theta - 1]$$

Maximum $1/r$ (minimum r) occurs at $\tilde\theta = 0$: $\frac{1}{r_m} = \frac{mZZ'e^2}{\ell^2}[\epsilon-1]$.

The apsidal angle is $\tilde\theta = 0$; we need $\psi = \tilde\theta(r\to\infty) = \tilde\theta(1/r\to 0)$:

$$0 = \epsilon\cos\psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos\psi = \cos\left(\frac{\pi-\theta}{2}\right) = \cos\frac{\pi}{2}\cos\frac{\theta}{2} + \sin\frac{\pi}{2}\sin\frac{\theta}{2} = \sin\frac{\theta}{2}$$

(using the repulsive-case relation $\theta = \pi - 2\psi \Rightarrow \psi = (\pi-\theta)/2$).

2.3 Attractive Case

Here $k > 0$, so $C_1 = km/\ell^2 \geq 0$. Take $\tilde\theta' = 0$, so at $\tilde\theta = \tilde\theta' = 0$: $1/r = C_1[1+\epsilon\cos(0)] = C_1[1+\epsilon]$. The apsidal point is also at $\tilde\theta = 0$; we again need $\psi = \tilde\theta(r\to\infty) = \tilde\theta(1/r\to 0)$.

$$\text{Maximum } 1/r \text{ (minimum } r\text{) is at } \tilde\theta = 0 : \frac{1}{r_m} = C_1[1+\epsilon]$$
$$0 = 1+\epsilon\cos\psi \Leftrightarrow -\frac{1}{\epsilon} = \cos\psi = \cos\left(\frac{\pi+\theta}{2}\right) = \cos\frac{\pi}{2}\cos\frac{\theta}{2} - \sin\frac{\pi}{2}\sin\frac{\theta}{2} = -\sin\frac{\theta}{2}$$

(using the attractive-case relation $\theta = 2\psi - \pi \Rightarrow \psi = (\pi+\theta)/2$).

3 Combining Both Cases: $S(\theta, E)$ and the Rutherford Cross Section

3.1 The Impact Parameter $S(\theta, E)$

Both cases give $\cos\psi = \pm\sin(\theta/2)$, i.e. $\cot^2(\theta/2) = 1/\sin^2(\theta/2) - 1$. Combining with $\cos\psi = 1/\epsilon$ (repulsive) or $-1/\epsilon = \cos\psi$ (attractive), in both cases $\cos^2\psi = 1/\epsilon^2$, and $\sin^2(\theta/2) = 1/\epsilon^2$ gives

$$\cot^2\frac{\theta}{2} = \frac{1}{\tan^2\frac{\theta}{2}} = \frac{\cos^2\theta/2}{\sin^2\theta/2} = \frac{1-\sin^2\theta/2}{\sin^2\theta/2} = \frac{1}{\sin^2\theta/2} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2}\right)^2$$

so

$$s = S(\theta, E) = \left|\frac{ZZ'e^2}{2E}\right|\cot\frac{\theta}{2} \tag{6}$$

3.2 The Rutherford Differential Cross Section

$$\frac{ds}{d\theta} = \left|\frac{ZZ'e^2}{2E}\right|\frac{d}{d\theta}\left(\cot\frac{\theta}{2}\right) = \left|\frac{ZZ'e^2}{2E}\right|\left(-\frac{1}{2}\frac{1}{\sin^2\theta/2}\right) = -\left|\frac{ZZ'e^2}{4E}\right|\frac{1}{\sin^2\theta/2}$$

Then

$$\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right| = \frac{1}{2}\left(\frac{ZZ'e^2}{2E}\right)^2 \frac{1}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} \cdot \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} \cdot \frac{1}{\sin^2\frac{\theta}{2}},$$

which simplifies to the Rutherford scattering cross section (for Coulomb forces):

$$\sigma(\theta) = \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \csc^4\left(\frac{\theta}{2}\right) \tag{7}$$

3.3 Remarks

1. The cross section is the same for the repulsive and attractive cases, since it is proportional to $(ZZ')^2$.
2. It diverges like E^{-2} at low energies.
3. It diverges like θ^{-4} (very strong!) for small angles θ.
4. This classical result is exactly the same as the result obtained in Quantum Mechanics.

3.4 Total Scattering Cross Section

$$\sigma_T = \int \sigma(\Omega)\, d\Omega$$

Remark 2. Notation: Goldstein writes $\sigma(\Omega) \leftrightarrow d\sigma/d\Omega$ (differential scattering cross section) and $\sigma_T \leftrightarrow \sigma$ (total scattering cross section); other texts may use different symbols for the same quantities.

In this case $\sigma(\theta) = \text{const} \times \frac{1}{\sin^4(\theta/2)}$, and $d\Omega = 2\pi\sin\theta\, d\theta = 4\pi\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta$, so

$$\begin{aligned} \sigma_T &= 4\pi \cdot \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}}\, d\theta \\ &\sim \int_0^\delta \frac{d\theta}{(\theta/2)^3} = 8\int_0^\delta \frac{d\theta}{\theta^3} = 8\frac{\theta^{-2}}{-2}\bigg|_0^\delta = +\infty \qquad \text{(strongly divergent)} \end{aligned}$$

Remark 3. Classically, $\sigma_T = +\infty$ for any long-range potential. In Quantum Mechanics, $\sigma_T = +\infty$ only for those long-range potentials $V(r)$ that do not decay fast enough as $r \to \infty$.

3.5 Caveat: Screened Coulomb (Yukawa) Potential

In a dense conducting material, the bare Coulomb potential $V = -k/r$ is screened:

$$V(r) = -\frac{k}{r} \longrightarrow V^{\text{screen}}(r) = -\frac{k}{r}e^{-r/\lambda} \qquad \text{(screened Coulomb potential, “Yukawa”).}$$

(In non-conducting materials $V^{\text{screen}}(r)$ may take a different form, but will still vanish much faster than $1/r$.)

4 Chapter 8: Hamiltonian and Hamilton’s Equations of Motion

4.1 Lagrangian and the Euler–Lagrange Equations

$$\begin{gathered} L = T - V = L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t) \\ 0 = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\bigg|_{q_j,\dot{q}_j} - \frac{\partial L}{\partial q_i}\bigg|_{q_j,\dot{q}_j}, \quad p_i = \frac{\partial L}{\partial \dot{q}_i}\bigg|_{q_j,\dot{q}_j}. \end{gathered}$$

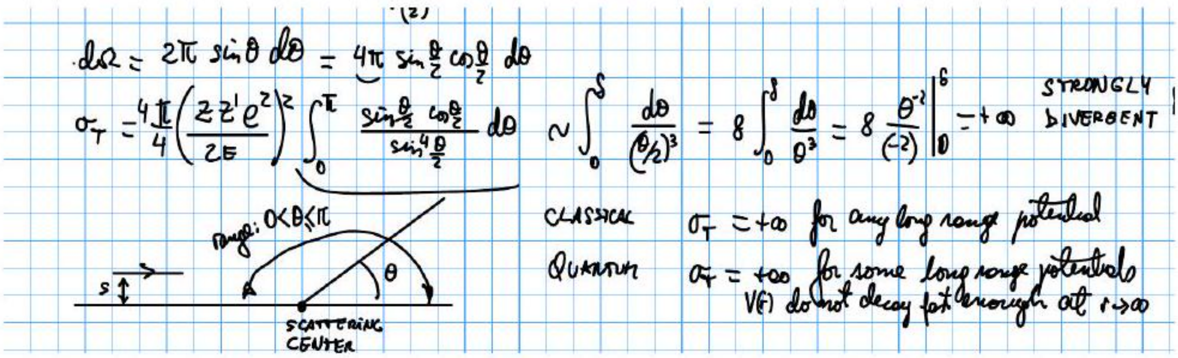

Figure 3: Total-cross-section divergence: the small-angle behavior of $\sigma(\theta) \propto \sin^{-4}(\theta/2)$ makes σ_T diverge, with the scattering-center geometry (impact parameter s, scattering angle θ) shown inset.

Remark 4. Midterm remark (marked invalid in the original notes, crossed out): in polar coordinates one might be tempted to write $\partial L/\partial\dot{\theta} = p_\theta = \ell_z$ and claim $L = L\left(r, \dot{r}, p_\theta\right)$, treating p_θ as an independent variable of the Lagrangian and applying the Lagrange equations to it directly. This is invalid — L is a function of coordinates and velocities $(q_i, \dot{q}_i)$, not momenta; conflating p_θ with a Lagrangian argument is exactly the error that the Legendre transformation to the Hamiltonian (below) is needed to correctly resolve. The professor crossed this out in the original notes as an invalid claim, rather than erasing it, precisely to flag it as a common mistake.

4.2 Legendre Transformation: a Thermodynamic Analogy

The Legendre transformation exchanges the role of a variable with its conjugate "slope" variable: for $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \to q_1, \dots, q_n, p_1, \dots, p_n$, recall for a function of two variables $f(x,y)$: $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$.

In Thermodynamics (fluid at constant particle number N), the First Law reads

$$dU = \delta Q + \delta W = TdS - pdV, \quad U = U(S,V) \text{ (internal energy).}$$

If instead we want (T,V) as the natural variables, define the Helmholtz free energy

$$F = A = U - TS$$

so that

$$dA = dU - d(TS) = (TdS - pdV) - TdS - SdT = -SdT - pdV \quad \Longrightarrow \quad A = A(T,V)$$

4.3 General Legendre Transformation

For a general $f(x,y)$ with $df = u\,dx + v\,dy,\ u = \partial f/\partial x,\ v = \partial f/\partial y$, to swap $(x,y) \to (u,y)$ define

$$g = f - ux \quad \Longrightarrow \quad dg = df - d(ux) = (u\,dx + v\,dy) - u\,dx - x\,du = -x\,du + v\,dy$$

so $g = g(u,y)$.

4.4 The Hamiltonian and Canonical Equations of Motion

Apply this to $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$ with $p_i = \frac{\partial L}{\partial \dot{q}_i}\Big|_{q_j, \dot{q}_j}$ playing the role of the conjugate variable to $\dot{q}_i$. Define

$$H = -\left(L - \sum_i p_i \dot{q}_i\right)$$

Then

$$dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i\right) - \frac{\partial L}{\partial t} dt + \sum_i (p_i d\dot{q}_i + \dot{q}_i dp_i)$$

and using $p_i = \partial L / \partial \dot{q}_i$ the $d\dot{q}_i$ terms cancel, leaving

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i\right) \quad \Longrightarrow \quad H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

Comparing coefficients of dp_i, dq_i, dt and using the Euler–Lagrange equations ($\partial L / \partial q_i = \dot{p}_i$) gives the canonical equations of motion:

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \tag{8}$$